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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: FUNDAMENTALS OF DATA ENGINEERING– (24DEA3101)
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TITLE OF THE PROJECT		
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Abstract:

Stock market data is generated continuously from various financial sources and contains valuable information for financial analysis and decision-making. However, collecting and processing this data manually can be time-consuming and may result in missing values, duplicate records, inconsistent formats, and data-quality issues. Managing large volumes of historical and frequently updated stock data using traditional methods is also difficult. Therefore, there is a need for an automated and reliable data engineering pipeline for efficient stock-market data processing.

To address these challenges, this project proposes a **Stock Market Data Pipeline** that automatically collects stock-market data, validates and cleans the incoming data, removes duplicate records, performs data transformation, and stores both raw and processed datasets efficiently. **Python** is used as the primary programming language for data ingestion, cleaning, validation, and transformation. The pipeline can use **PostgreSQL and Parquet** for structured

and efficient data storage, while technologies such as **Apache Kafka and PySpark** can be integrated for real-time streaming and large-scale data processing.

The proposed pipeline follows a systematic workflow consisting of data collection, raw-data storage, validation, cleaning, duplicate removal, transformation, analytical processing, and visualization. It can generate useful analytical metrics such as **price changes, returns, moving averages, volume metrics, and volatility**. The pipeline also supports separate raw, cleaned, and curated data layers, improving data quality, reliability, and maintainability.

Overall, the Stock Market Data Pipeline provides an automated and scalable solution for converting raw stock-market data into **clean, reliable, and analysis-ready datasets**. The project demonstrates important Fundamentals of Data Engineering concepts such as **ETL, data validation, data transformation, batch processing, data storage, stream processing, analytics, and visualization**. In the future, the system can be extended with cloud deployment, real-time market feeds, automated monitoring, anomaly detection, and machine-learning-based analytics.